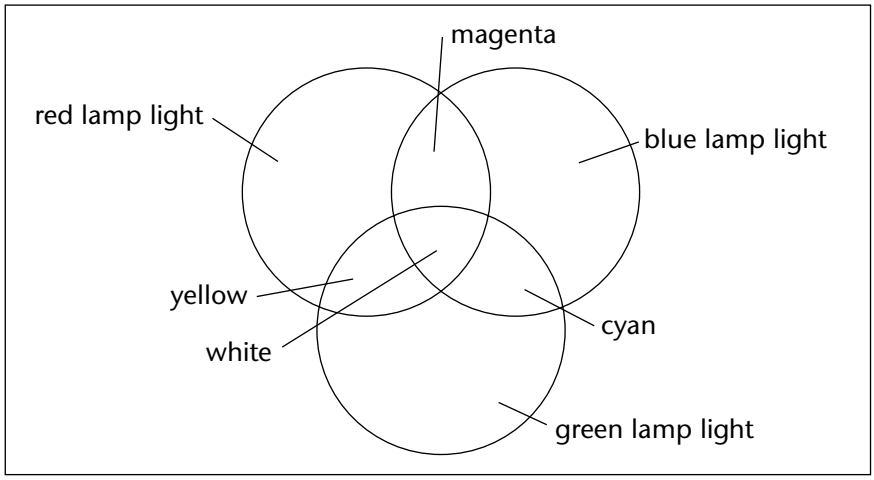
Whose colour?

11a

1 Cross out the incorrect term from each set of brackets.

Let the stage light operator speak:

Lights of various (colours / colors) are shone onto the stage to (creat / create) certain (affects / effects). Often coloured (lights / lites) are mixed by (shining / shineing) them on the same (spot / spots). When red and (blu / blue) lights are mixed, a magenta colour, which is dark red, is (produced / producted); green and red make (yello / yellow); and blue and green give cyan (light blue); and all (free / three) colours (to gether / together) produce white light.

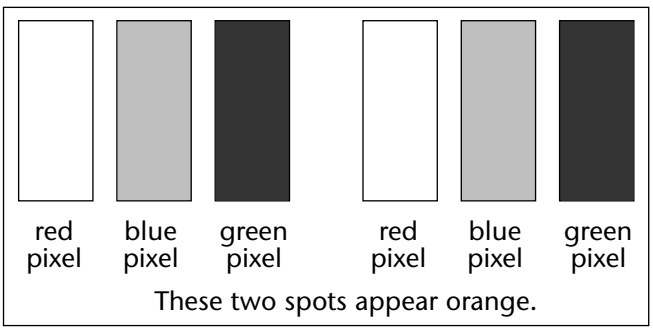


2 Cross out the incorrect tense from each set of brackets.

Let the television engineer speak:

Television screens and other colour monitors (are made up / were made up of) of thousands or even millions of pixels, tiny coloured dots or stripes. These pixels (are stimulated / will be stimulated) and then (glow / glowed) for a very short time. Three primary colours (are used / were used): red, green and blue. From these all other colours (can be produced / have been produced)). Three separate electron guns in a TV picture tube (are stimulating /stimulate) each colour pixel. Flat screens (had /have) bubbles which (are stimulated / were stimulated). Because the coloured dots (are / were) too small to be seen individually, they (merge / merged) to produce one mixed colour spot. If a group of red and green dots (are lit up / will be lit up), yellowish spots (result / results). But if the green dots (are / will be) not so bright, the coloured spots (were appeared / would appear) orange.

You (have been able to see / might be able to see) the colour pixels, if you (look / will look) at a TV screen from very close. A computer monitor (had / has) much smaller pixels (having resulted / resulting) in a sharper picture.



3 Cross out the incorrect preposition from each set of brackets.

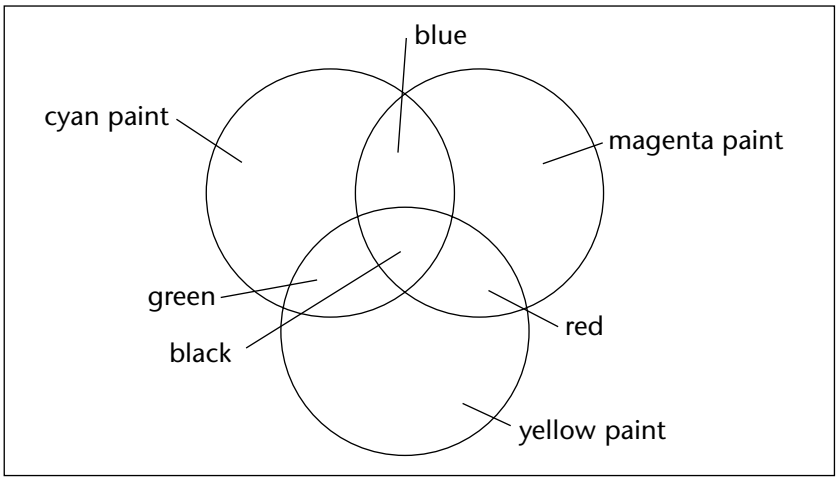
Let the painter speak:

A painter doesn't use light (from / of) dots or lamps. A painter mixes coloured pastes and relies (by / on) the colours which are reflected (for / from) those pastes. Magenta paint mixed (on / with) yellow paint gives a red colour; and cyan and magenta produce a blue colour. All colours mixed together end up black. The three paints cyan, magenta and yellow produce the greatest range of colours when mixed.

See diagram, next page.

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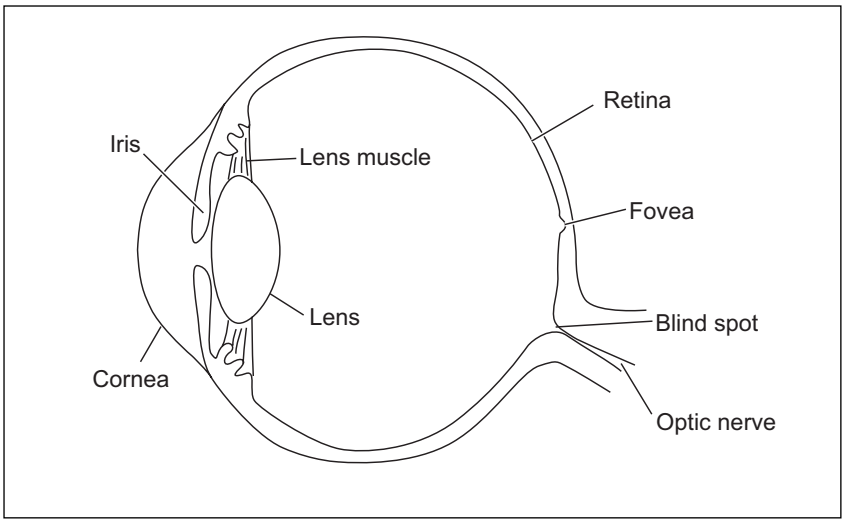
Whose colour?



A painter uses colour to create mood, shape, perspective, depth, contrast and harmony. Very subtle changes (in / on) hue and colour result (in / on) interesting effects.

4 Order the sentences in each of the three paragraphs by writing numbers in front of the sentences.

Let the doctor speak:



Paragraph 1:

☐ **a** Light stimulates nerves in the eye that send a message to the brain.

☐ **b** Then the brain produces a picture.

Paragraph 2:

☐ a On the retina light sensitive nerve endings are stimulated: When it is dark, rods are stimulated to produce a black and white picture; when it is bright, three types of cones (i.e. red, blue and green cones) are stimulated to varying degrees and produce a colour picture in our brain.

☐ **b** The brain interprets them and builds up a picture.

☐ **c** The stimulated nerve endings, i.e. the rods and cones, send electric impulses to the brain.

☐ **d** When light enters the eye, the cornea and lens focus the light rays onto the retina (usually near the spot called the fovea).

(continued next page)

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Whose colour?

Paragraph 3:

- ☐ **a** Because the brain and not the eye develops the picture, it is possible to see something which is not even there.
- ☐ **b** However, despite the existence of your blind spot, when you normally look around, there is no hole or gap in the pictures that you see.
- ☐ **c** It is the brain that compensates for the hole due to the blind spot and so completes the image in the brain.
- ☐ **d** Right then the image of the 'X' falls on the blind spot in your eye (there are no rods and cones where the optic nerve leaves the eye ball) and so you cannot see the 'X'.
- ☐ **e** When you look at the dot below with only your left eye and move the paper slowly backwards and forwards, at some point (when the page is about 15 cm from your eye) you will no longer be able to see the 'X'.
- ☐ **f** X ●

5 Summarise the information by completing the table as much as possible.

Let the biologist speak:

It is difficult to know exactly what other animals see. Many animals do not see like humans, who have three types of cones in their retina. Most insects, birds and reptiles have much better vision than humans. Bees see ultra-violet as well as blue and green light, so flowers would look quite different to them. Many birds have four types of cones; the fourth one picks up ultra-violet light. Pigeons have five types of cones and therefore are able to distinguish better between different 'colours'. Snakes see colour as well as infra-red, so they can perceive heat from other animals. Dogs have really poor colour vision. Cows see only black, white and shades of grey. Sharks have no colour vision either. Squirrels are red-green colour blind. Many simpler animals such as snails and jellyfish can only distinguish between light and dark; their 'pictures' of the surrounding are very crude.

Animal	Number of cone type	Black and white vision?	Colour vision?	Other vision?	Vision better or worse than human?

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Whose colour?

6 Write a report on light. Use the details below and other knowledge you have gained from this worksheet. First write a title. Then order the information into three paragraphs. In the first paragraph write about electromagnetic radiation in general. In the second paragraph, write about light. In the third paragraph, provide some historical information.

Let the scientist speak:

- Part of electromagnetic (EM) spectrum
- EM waves carry energy, travel at 300 000 000 m/s, can travel in vacuum, can travel through some media.
- Distinguished by wavelength, frequency, energy
- Main EM types in order of wavelength (long to short), frequency (low to high) and energy (low to high): radio waves, microwaves, infra-red waves, light, ultra-violet rays, X-rays, gamma rays.
- Light: Red, orange, yellow, green, blue, indigo, violet
- Wavelengths: red light: 700 nm (nanometre = 0.000 000 001 metre = 10^{-9} m); violet light: 400 nm
- Newton (1672) showed white light is a mixture of rainbow colours – a triangular prism splits white light into roygbiv; roygbiv colours combine into white light.

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